

Performance Analysis of SRF PLL Type-3 with Dynamic Feed Forward Frequency Estimator for Abnormal Grid Conditions

Seema Agrawal^{1,✉}, Mahendra Kumar², and D K Palwalia¹

¹Rajasthan Technical University Kota

²Indian Institute of Technology Roorkee

³RTU Kota

✉sagrawal@rtu.ac.in

Abstract

A phase-locked loop (PLL) is a closed-loop feedback control device that synchronizes its output signal with an input signal in both frequency and time. This paper presents a robust method for tracking against adverse conditions of the fundamental sequence component of utility voltage. The proposed phase-locked loop (PLL) is a hybrid configuration of the phase-locked loop type-3 (SRF type-3) synchronous reference frame coupled with a feed-forward frequency estimator using a selective harmonic filtering technique. A SRF type -2 based PLL has a secondary closed path. Under the ramp frequency transition, it has a steady-state step and a frequency error. In the proposed PLL, the constant gain loop filter is used to eliminate the phase and frequency error. Feed-forward operation with selective harmonic pre-filtering enables fast-tracking with a low tracking error of the reference signal. From the results it is evident that the proposed PLL achieves a high bandwidth and quick dynamic response without endangering the stability and filtering capability. The proposed system has been tested through MATLAB Simulink platform under abnormal conditions.

Keywords: Frequency estimation, Phase-locked loop (PLL), Secondary closed path (SCP), Synchronous reference frame (SRF), Transient response.

1 Introduction

Identification of the fundamental sequence components of utility voltage operating under adverse conditions is a crucial factor in regulating three-phase grid-connected power converters [1]. Positive sequence voltage detection at the fundamental frequency is critical for controlling distributed generation, storage network, active power filter, power factor control, power line conditioner, and other grid-connected power conditioning equipment [2].

A phase-locked loop (PLL) is a closed-loop feedback control device that synchronizes its output signal with an input signal in both frequency and time. PLL structure consists of a phase detector (PD), a loop filter (LF), and an oscillator regulated by voltage (VCO). The versatility of the structure makes the synchronization process the most commonly accepted approach for analog and digital signals of implementation [3].

Improved transient output without endangering the stability and filtering capability is a major challenge associated with PLL. There have been numerous design aspects reported in the literature to overcome this issue [4]; [5]; [6]. Synchronization methods such as zero-crossing detection, stationary reference frame, and synchronous reference frame (SRF)PLL are pursued usefulness. A detection method of zero-crossing is simple, but the zero-crossing point is susceptible to harmonics and notches. In addition, a stationary PLL-based reference frame structure cannot reliably assess the phase tracking under an unbalanced voltage condition [7]. After that point, due to the double frequency ripple in the error signal, the synchronous reference frame dependent PLL is ineffective under the unbalance of the three-phase input signal.

Techniques such as placing additional filters within the loop, pre-filtering before the phase detector, and implementing complex structures have been suggested to enhance SRF-PLL's transient response and filtering [8]; [9]; [10]. With phase unbalancing and harmonics the use of extra filters in SRF-PLL, such as notch and moving average, removes high-frequency ripples [11]; [12]. To manage the unbalanced state, a decoupled double SRF PLL is implemented with two synchronously rotating frames of opposite angular velocity. It is capable of monitoring variability in frequencies, but needs some refinement under harmonic conditions. The PLL based on the program is used in a series-connected converter that uses the technique of delayed cancellation [13]. An active power filter plays a major role in the quality of power, mitigating har-